



Approximation Algorithms for Big Data

Lecture 9: Approximate Regression and Low Rank Approximation

Spring 2026

Instructor: Tal Wagner

Administrative Announcements

- HW2 grades published
- HW3 (last one) out tonight, due July 16 (day of final class)

Administrative Announcements

- University-wide teaching survey is live:
- https://my.tau.ac.il/TAU_Student/TeachingSurvey
- Your feedback is highly valuable

הדקנט לחדשנות
בהוראה ובלמידה
אוניברסיטת תל אביב



סקר שביעות רצון מהוראה
סמסטר ב', תשפ"ז

יפתח מחר, 23/6 בשעה 8:00 למענה

ספרו לנו, איך עבר הסמסטר?



מקנה
נקודות
בבידינג

הקליקו לסקר

כניסה למידע האישי > מענה על הסקר

הסקר יימשך שלושה וחצי שבועות
ויהיה פתוח למענה עד ל-18/7/26

Administrative Announcements

- July 9 (penultimate class): Guest lecture by **Dori Eldar** from **Mobileye**
- **Topic:** probabilistic framework for aligning multiple vehicle drives over a local map
- Details published on email and class forum on Moodle



The poster is for a "SPECIAL GUEST LECTURE" titled "AAA: Alignment, Assignment & Aggregation" with the subtitle "A Principled Approach to Multi-Drive Map Alignment". The speaker is Dori Eldar, Principal Engineer at Mobileye. The event is on July 9, 2026, at 15:00, in the Goldschlager Multidisciplinary Engineering Building, 3rd Floor, Room 315. Logos for Mobileye and Tel Aviv University are at the top.

mobileye™

אוניברסיטת תל אביב
TEL AVIV UNIVERSITY

SPECIAL GUEST LECTURE

AAA: Alignment, Assignment
& Aggregation

A Principled Approach to Multi-Drive Map Alignment

 Dori Eldar
Principal Engineer, Mobileye

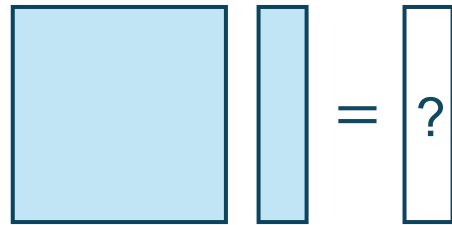
July 9, 2026 | 15:00
Goldschlager Multidisciplinary Engineering Building
3rd Floor, Room 315

Plan

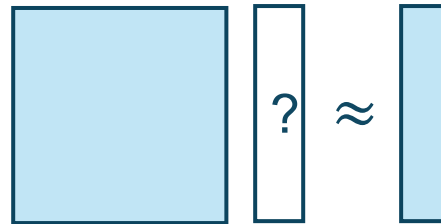
- Recap last time (with corrections)
- Today's topic 1: Approximate regression via sketching
- Today's topic 2: Approximate low-rank approximation via sketching

Computational Linear Algebra

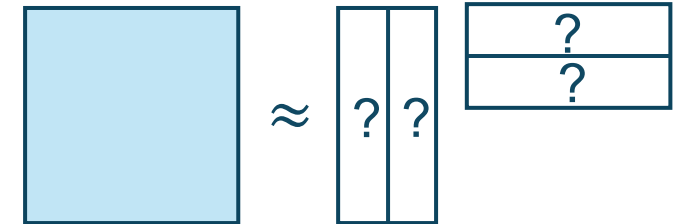
- Extremely useful and prevalent
- Extremely computationally taxing
- We will develop efficient randomized approximation algorithms in the spirit of class
- Why would the techniques we've seen so far be related?



Matrix-vector /
matrix-matrix
multiplication



Linear regression



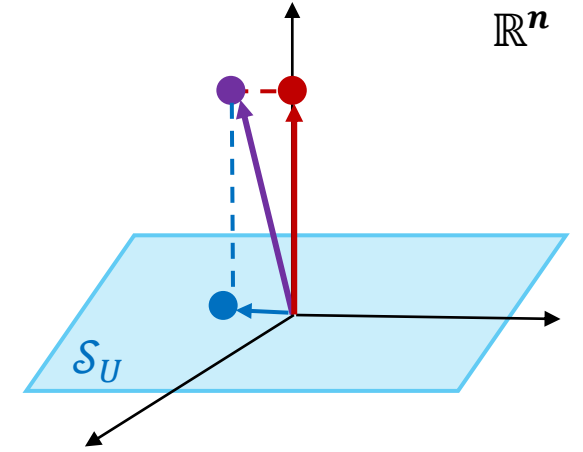
Low-rank
approximation

Linear Algebra Background

Matrix Norms

- Let $A \in \mathbb{R}^{n \times d}$
- The **Frobenius norm** is: $\|A\|_F := \sqrt{\sum_{i,j} A_{ij}^2}$
- The **spectral norm** is: $\|A\|_2 := \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$
- Fact: For every $A \in \mathbb{R}^{n \times d}$ and $B \in \mathbb{R}^{d \times k}$, we have $\|AB\|_F \leq \|A\|_2 \|B\|_F$.
 - **Proof?**

Orthogonal Projections



- Let $U \in \mathbb{R}^{n \times d}$ have **orthonormal columns** ($U^T U = I_{d \times d}$)
- Its columns form a basis of a **d -dimensional subspace** $\mathcal{S}_U$ of $\mathbb{R}^n$
- How do we project a vector onto $\mathcal{S}_U$?
 - The orthogonal projection matrix **onto** $\mathcal{S}_U$ is UU^T
 - Thus: the orthogonal projection of $x \in \mathbb{R}^n$ onto $\mathcal{S}_U$ is $UU^T x$
 - The orthogonal projection matrix **against** $\mathcal{S}_U$ is $I - UU^T$
 - $x = UU^T x + (I - UU^T)x$ is the corresponding direct sum decomposition
- **The Pythagorean identity:** $\|x\|_2^2 = \|UU^T x\|_2^2 + \|(I - UU^T)x\|_2^2$ (proof?)
- **Extended to matrices** $M \in \mathbb{R}^{n \times m}$: $\|M\|_F^2 = \|UU^T M\|_F^2 + \|(I - UU^T)M\|_F^2$ (proof?)

Singular Value Decomposition (SVD)

- Every $A \in \mathbb{R}^{n \times d}$ can be written as $A = U\Sigma V^T$ such that:

- $U \in \mathbb{R}^{n \times k}$ has orthonormal columns ($U^T U = I_{k \times k}$)
- $V \in \mathbb{R}^{d \times k}$ has orthonormal columns ($V^T V = I_{k \times k}$)
- $\Sigma \in \mathbb{R}^{k \times k}$ is diagonal with positive entries $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$
- k is the rank of A

$$A = U \Sigma V^T$$

- Equivalent writing: $A = \sum_{i=1}^k \sigma_i \cdot u_i v_i^T$ (A as the sum of r rank-1 matrices)
- The σ_i are the **singular values** of A
 - The columns of U are called **left-singular vectors**
 - The columns of V are called **right-singular vectors**

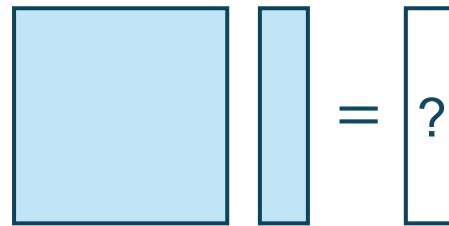
Singular Values and Matrix Norms

- Let $A \in \mathbb{R}^{n \times d}$ have SVD $A = U\Sigma V^T = \sum_{i=1}^r \sigma_i \cdot u_i v_i^T$.
- Fact 1: The spectral norm is the top singular value: $\|A\|_2 = \sigma_1$.
 - Proof?
- Fact 2: The Frobenius norm is the ℓ_2 -norm of singular values: $\|A\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2}$.
 - Proof?
 - Use two properties that are easy to verify by direct calculation:
 1. $\|A\|_F^2 = \text{trace}(A^T A)$
 2. $\text{trace}(BC) = \text{trace}(CB)$ for any two matrices B, C (“trace is cyclic”)

Moore-Penrose Pseudo-Inverse

- The **Moore-Penrose pseudo-inverse** A^+ of a matrix $A \in \mathbb{R}^{n \times d}$ is a generalization of the matrix inverse that exists (uniquely) for **every** matrix $A \in \mathbb{R}^{n \times d}$
 - Even if A is square with non-full rank
 - Even if A is rectangular
- **Definition:**
 - Write A in SVD form $A = U\Sigma V^T = \sum_{i=1}^r \sigma_i \cdot u_i v_i^T$
 - The pseudo-inverse is $A^+ := \sum_{i=1}^r \sigma_i^{-1} \cdot u_i v_i^T$
- **Properties:** A^+ satisfies:
$$AA^+A = A; \quad A^+AA^+ = A^+; \quad (AA^+)^T = AA^+; \quad (A^+A)^T = A^+A$$
- A^+ is the unique matrix that satisfies all four properties

Approximate Matrix Multiplication



Approximate Matrix Multiplication (AMM)

- Let $A \in \mathbb{R}^{n \times d}$ and $B \in \mathbb{R}^{n \times k}$
- What is the time for computing $A^T B \in \mathbb{R}^{d \times k}$ exactly?
- Definition: An (ε, δ) -AMM algorithm returns a matrix $C \in \mathbb{R}^{d \times k}$ that satisfies

$$\Pr[\|A^T B - C\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- Approach – “**Sketch and Solve**”:
 - Sample a random sketching matrix $S \in \mathbb{R}^{m \times n}$
 - **Sketch**: compute SA and SB
 - **Solve**: return $C = (SA)^T (SB)$ as the approximation for $A^T B$
 - Which random sketching matrix can we use?

Recap: JL

- **DJL Theorem:** Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $S \in \mathbb{R}^{m \times n}$ be a matrix with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$. For every vector $z \in \mathbb{R}^d$ with unit norm $\|x\|_2 = 1$, we have

$$\Pr[1 - \varepsilon \leq \|Sx\|_2 \leq 1 + \varepsilon] \geq 1 - \delta$$

- **JL Theorem:** For every given set of N vectors $x_1, \dots, x_N \in \mathbb{R}^n$, if $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$

$$\Pr\left[\forall i, j: \|Sx_i - Sx_j\|_2 = (1 \pm \varepsilon)\|x_i - x_j\|_2\right] \geq 1 - \delta$$

- **Theorem – AMM via JL:**

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- What is the running time?
- Do you think it can be improved? How?

The JL Moment Property (JLMP) with corrections

- Definition: a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ satisfies the (ε, δ) -second moment JL property (2-JLMP) if for every unit vector $x \in \mathbb{R}^n$:

$$\sqrt{\mathbb{E}[(\|Sx\|_2^2 - 1)^2]} \leq \varepsilon\sqrt{\delta}$$

- Note this implies: $\sqrt{\mathbb{E}[\|Sx\|_2^2 - \|x\|_2^2]} \leq \varepsilon\sqrt{\delta}\|x\|_2$ for every $x \in \mathbb{R}^n$
- **Claim**: S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2\delta$ satisfied 2-JLMP
- **Proof**: Let $x \in \mathbb{R}^n$ be a unit vector.

- For every $j = 1, \dots, m$ we have: $\mathbb{E}[(Sx)_j^2] = \mathbb{E}\left[\left(\sum_{i=1}^n S_{ji}x_i\right)^2\right] = \mathbb{E}\left[N\left(0, \frac{1}{m}\|x\|_2^2\right)^2\right] = \frac{1}{m}$
- Hence: $\mathbb{E}[\|Sx\|_2^2] = \mathbb{E}\left[\sum_{j=1}^m (Sx)_j^2\right] = 1$ 

$$\text{Hence } \mathbb{E}[(\|Sx\|_2^2 - 1)^2] = \text{Var}(\|Sx\|_2^2) = \text{Var}\left(\sum_{j=1}^m N\left(0, \frac{1}{m}\|x\|_2^2\right)^2\right) = \frac{1}{m^2} \sum_{j=1}^m \text{Var}(N(0,1)^2) = \frac{2}{m}.$$

2-JLMP:

From JLMP to AMM (1/2)

$$\forall x \in \mathbb{R}^n: \sqrt{\mathbb{E}[\|Sx\|_2^2 - \|x\|_2^2]} \leq \varepsilon \sqrt{\delta} \|x\|_2^2$$

- **Lemma:** Let $x, y \in \mathbb{R}^n$ be unit vectors.. If S satisfies 2-JLMP then:

$$\sqrt{\mathbb{E}[x^T S^T S y - x^T y]^2} \leq \varepsilon \sqrt{\delta}$$

- **Proof:**

- We have the identity: $x^T y = \frac{1}{4} (\|x + y\|_2^2 - \|x - y\|_2^2)$, hence:

- $\sqrt{\mathbb{E}[x^T S^T S y - x^T y]^2}$

- $= \frac{1}{4} \sqrt{\mathbb{E} \left[\left((\|S(x + y)\|_2^2 - \|x + y\|_2^2) - (\|S(x - y)\|_2^2 - \|x - y\|_2^2) \right)^2 \right]}$

- $\leq \frac{1}{4} \sqrt{\mathbb{E}[(\|S(x + y)\|_2^2 - \|x + y\|_2^2)^2]} + \sqrt{\mathbb{E}[(\|S(x - y)\|_2^2 - \|x - y\|_2^2)^2]}$

- $\leq \frac{1}{4} \varepsilon \sqrt{\delta} (\|x + y\|_2^2 + \|x - y\|_2^2)$

- $\leq \frac{1}{4} \varepsilon \sqrt{\delta} (2\|x\|_2^2 + 2\|y\|_2^2)$

- $= \varepsilon \sqrt{\delta}$



Corollary:

$$\forall x, y \in \mathbb{R}^n: \mathbb{E}[x^T S^T S y - x^T y]^2 \leq \varepsilon \sqrt{\delta} \|x\|_2^2 \|y\|_2^2$$

Lemma from previous slide:

$$\forall x, y \in \mathbb{R}^n: \\ \mathbb{E}[x^T S^T S y - x^T y]^2 \leq \varepsilon \sqrt{\delta} \|x\|_2^2 \|y\|_2^2$$

From JLMP to AMM (2/2)

- **Lemma:** Let $A \in \mathbb{R}^{n \times d}$ and $B \in \mathbb{R}^{n \times k}$. If S satisfies 2-JLMP, then we have AMM:

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon \|A\|_F \|B\|_F] \geq 1 - \delta$$

- **Proof:** Let $a_1, \dots, a_d \in \mathbb{R}^n$ and $b_1, \dots, b_k \in \mathbb{R}^n$ be the columns of A and B .

- $\mathbb{E}[\|(SA)^T(SB) - A^T B\|_F^2] = \mathbb{E}\left[\sum_{i=1}^d \sum_{j=1}^k \left((Sa_i)^T(Sb_j) - a_i^T b_j\right)^2\right]$

- $= \sum_{i=1}^d \sum_{j=1}^k \mathbb{E}\left[(Sa_i)^T(Sb_j) - a_i^T b_j\right]^2$

- $\leq \varepsilon^2 \delta \sum_{i=1}^d \sum_{j=1}^k \|a_i\|_2^2 \|b_j\|_2^2$

- $= \varepsilon^2 \delta \|A\|_F^2 \|B\|_F^2$

- Hence by Markov: $\Pr[\|(SA)^T(SB) - A^T B\|_F^2 > \varepsilon^2 \|A\|_F^2 \|B\|_F^2] < \delta$. 

- We proved the AMM via JL theorem – what running time did we get?

AMM: Upshot

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- We proved:
 - S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfied 2-JLMP:

$$\sqrt{\mathbb{E}[(\|Sx\|_2^2 - 1)^2]} \leq \varepsilon\sqrt{\delta}$$

AMM: Upshot

- We proved:
 - S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfied 2-JLMP:

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- 2-JLMP implies AMM:

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon\|A\|_F\|B\|_F] \geq 1 - \delta$$

AMM: Upshot

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- Therefore: S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfies AMM

AMM: Upshot

- We proved:
 - S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfied 2-JLMP:

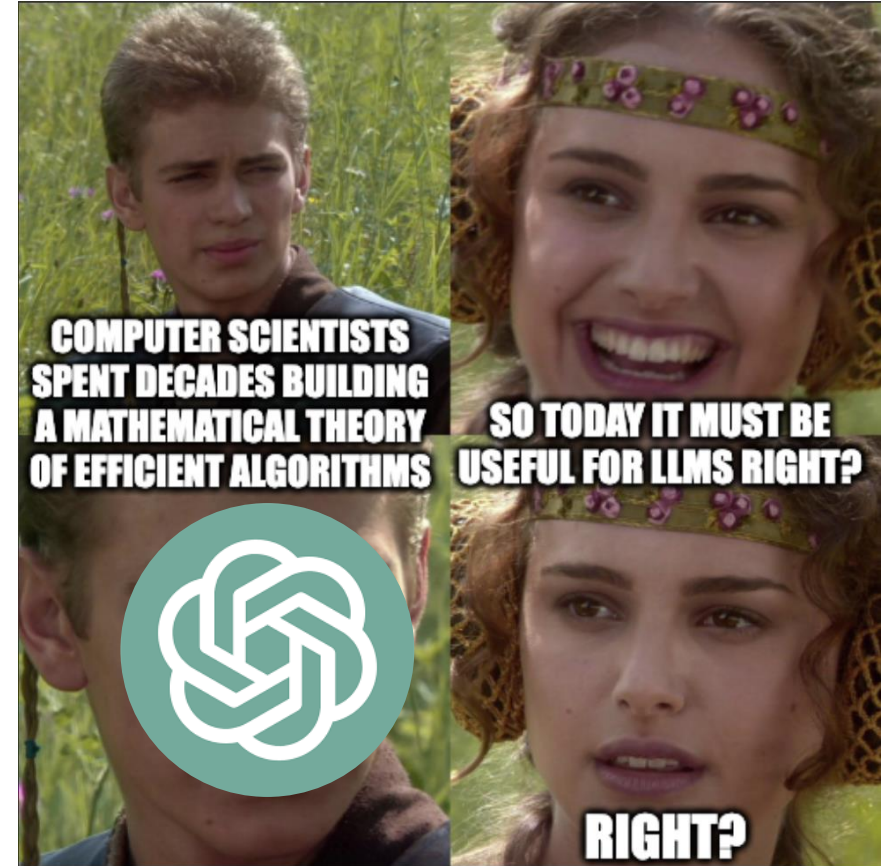
$$\sqrt{\mathbb{E}[(\|Sx\|_2^2 - 1)^2]} \leq \varepsilon\sqrt{\delta}$$

- 2-JLMP implies AMM:

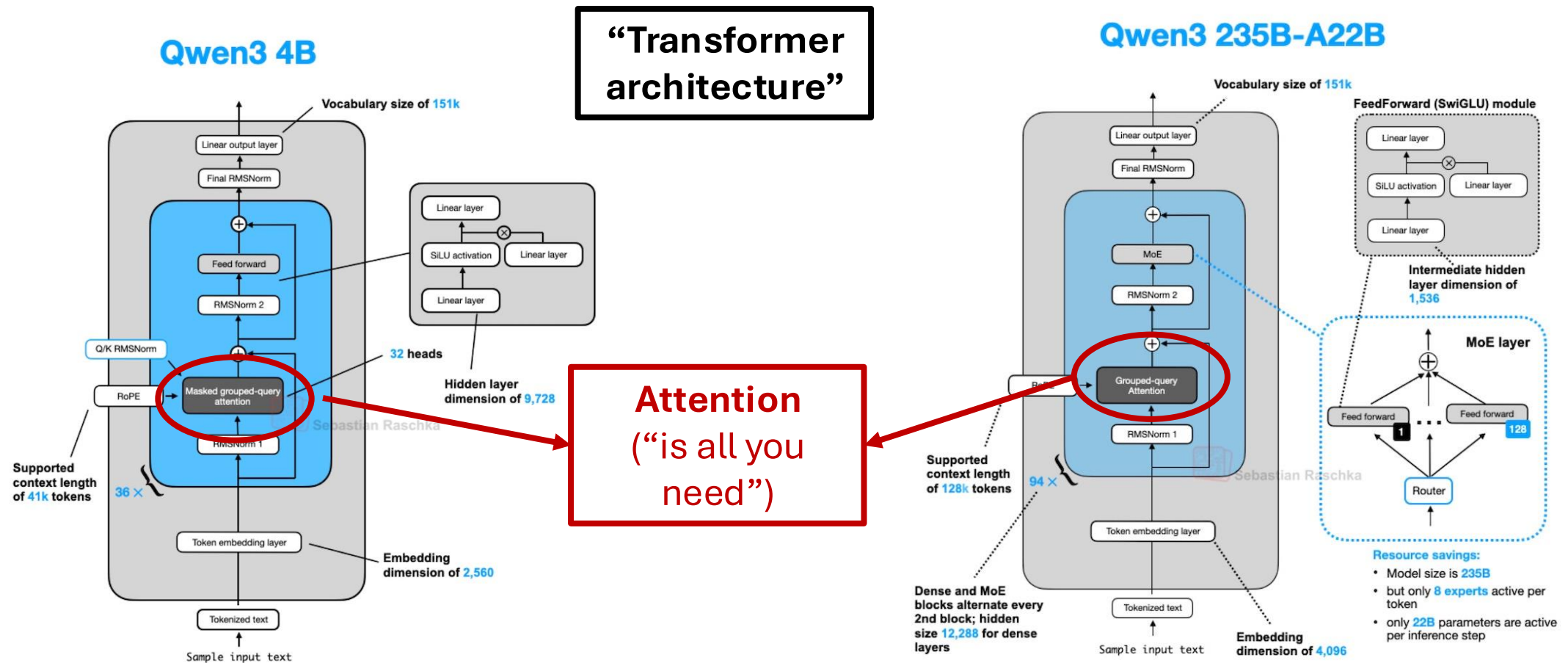
$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon\|A\|_F\|B\|_F] \geq 1 - \delta$$

- Therefore: S with iid entries from $\frac{1}{\sqrt{m}} N(0,1)$ with $m = 1/\varepsilon^2 \delta$ satisfies AMM
- Remark: Can get AMM with only $m = \log\left(\frac{1}{\delta}\right)/\varepsilon^2$
 - Can be proved via higher JL moments

Sketching-based approximation algorithms in LLMs and AI



Crash Course on Large Language Models



Attention

- **Definition – Softmax:**

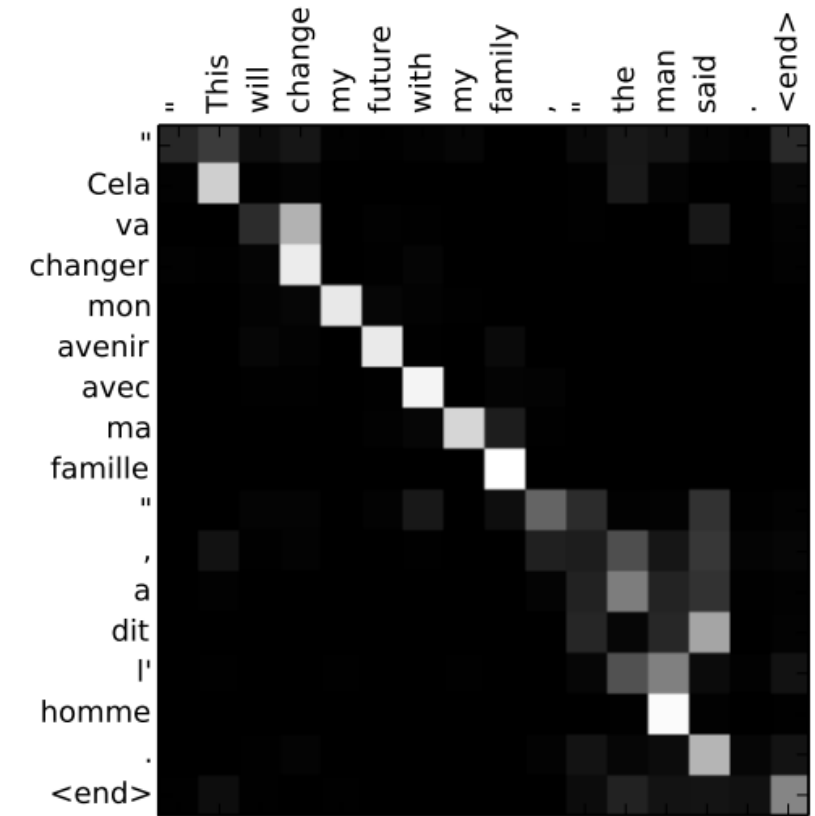
- The softmax operation $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is defined as: $\phi(x)_i = \frac{e^{x_i}}{\sum_{j=1}^d e^{x_j}}$
- Softmax on matrix $M \in \mathbb{R}^{n \times d}$ is applied separately on each row

- **Definition – the Attention operation:**

- Input: $K, Q, V \in \mathbb{R}^{n \times d}$ (their rows are called keys, queries, values)
- Output:
 - Compute $A := \phi\left(\frac{1}{\sqrt{d}} QK^T\right)$ (A is called the Attention matrix)
 - Return AV
- **Running time?**
- n is the “**context length**” – number of tokens the LLM needs to process at once
 - **major bottleneck in AI** – **usually requires efficient approximations**

Attention - Intuition

- The goal of the Attention matrix A is to describe which tokens are related to which tokens in the sequence
- How this works:
 - The LLM computes **high-quality embeddings of the tokens as vectors in $\mathbb{R}^d$** (keys, queries, values) such that if two tokens t_i, t_j are related, then their key-query inner product $q_i^T k_j$ is large
 - The matrix multiplication of the Attention matrix with the values, AV , computes even better embeddings for the next Attention based on the results of the current Attention
 - The final embeddings are used for next-token prediction



$$A := \phi \left(\frac{1}{\sqrt{d}} Q K^T \right)$$

Output: AV

Attention: Efficient Approximation?

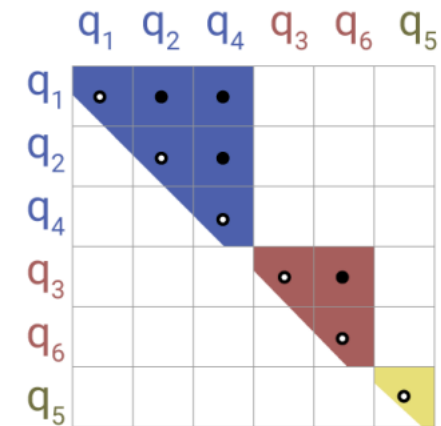
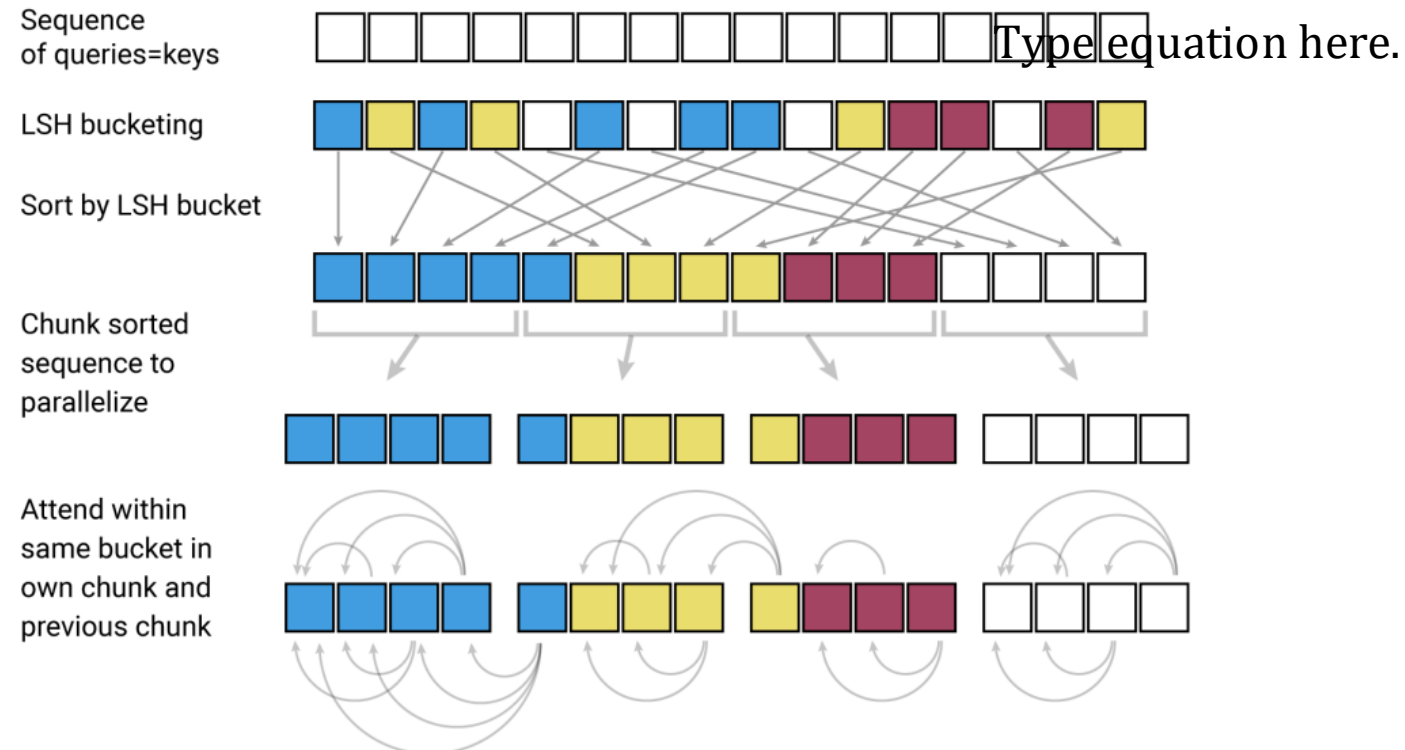
- **Upshot:**

- We have two sets of vectors: queries $q_1, \dots, q_n$ and keys $k_1, \dots, k_n$ in $\mathbb{R}^d$
 - We have a “softmax kernel”: $k(x, y) \propto e^{q_i^T k_j}$
 - The Attention matrix A is the corresponding kernel matrix
 - We want to compute the kernel matrix-vector product Av for some given v 's
 - The context length n is extremely large
-
- Have we seen techniques that could be relevant?
 - There are very many papers taking this approach; we'll see a small sample
 - **Disclosure:** this is very popular in research papers but less so in practice (why?)

“Reformer”: Attention via LSH

[Kitaev, Kaiser, Levskaya; ICLR 2020]

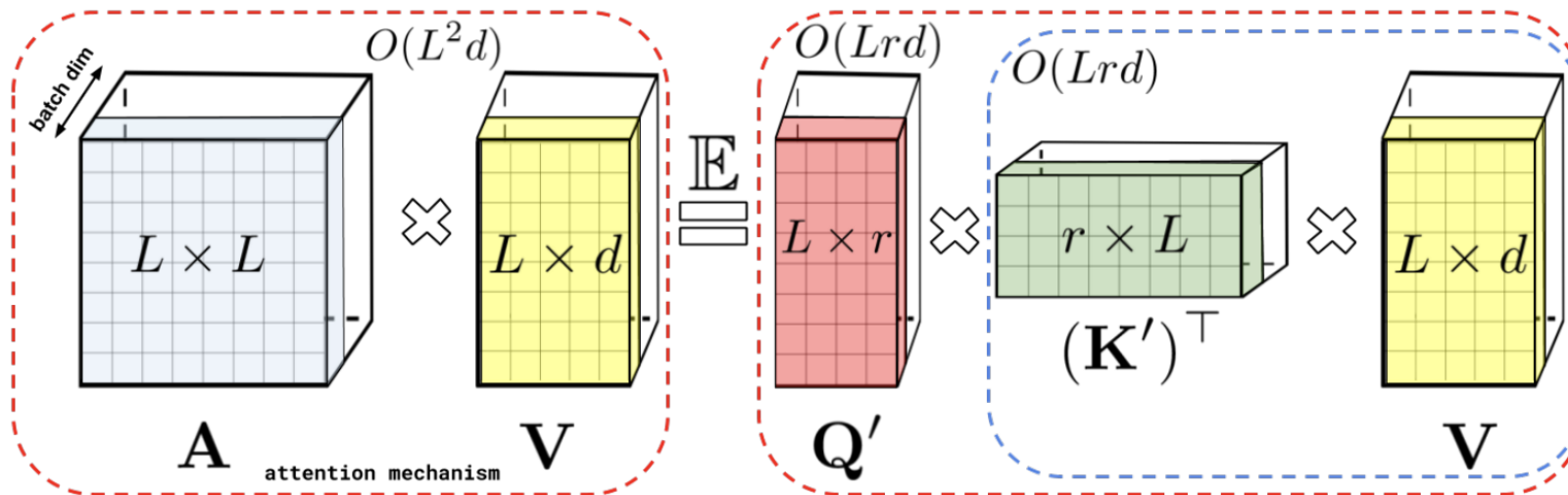
- We want to find query-key pairs with large inner product
- Use LSH!



“Performer”: Attention via Random Features

[Choromanski et al.; ICLR 2021]

- Attention is a kernel matrix
- Approximate it with random features!
- They develop *positive* random features because positive/negative random features like we’ve seen (e.g., RFF) lead to too large errors for near-zero values



KDEformer: Attention via Fast KDE

[Zanideh, Han, Daliri, Karbasi; ICML 2023]

- Attention is kernel matrix-vector multiplication
- Approximate it with fast KDE techniques like HBE and follow-ups!

HyperAttention: Attention via AMM

[Han, Jayaram, Karbasi, Mirrokni, Woodruff, Zanideh; ICLR 2024]

- Attention is matrix-vector multiplication
- Approximate it with sketch-and-solve approximate matrix multiplication!

Thinformer: Attention via KDE Coresets

[Carrell, Gong, Shetty, Dwivedi, Mackey; ICML 2025]

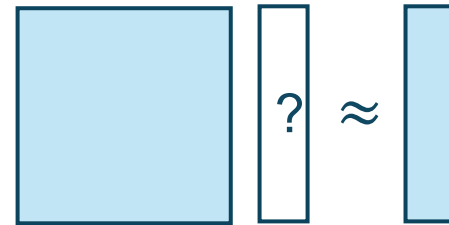
- Attention is a kernel matrix
- Approximate by sampling a small KDE-preserving subset (i.e., coreset!)
- Based on a recently published new “kernel thinning” technique for kernel coresets

RACE Attention: Attention via LSH Features

[Joshi, Chowdhury, Kanakamedala, Singh, Tu, Shrivastava; ICLR 2026]

- Attention is a kernel matrix
- Approximate it with LSH-based random features!
- The “softmax kernel” is not LSHable; they replace Attention with an LSHable analog

Regression



ℓ_2 -Regression

- Definition: in the ℓ_2 -regression problem,
 - Input: $A \in \mathbb{R}^{n \times d}$ and $b \in \mathbb{R}^d$
 - Output: $x^* = \min_{x \in \mathbb{R}^d} \|Ax - b\|_2$ (i.e., solve the linear equation as best as possible)

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 - Exact solution?

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 - **Exact solution?** $x^* = (A^T A)^+ A^T b$

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 - Time of exact solution?

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 - Exact solution? $x^* = (A^T A)^+ A^T b$
 - Time of exact solution?
- Approximate ℓ_2 -regression : Given ε, δ , find x such that

$$\Pr[\|Ax - b\|_2 \leq (1 + \varepsilon) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2] \geq 1 - \delta$$

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- Approximate ℓ_2 -regression : Given ε, δ , find x such that

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- Ideas how to achieve this?

Oblivious Subspace Embedding (OSE)

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- **Definition:** a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ is an (ε, δ, d) -oblivious subspace embedding (OSE) for $\mathbb{R}^n$ if for every given subspace $\mathcal{V} \subset \mathbb{R}^n$ of dimension d , we have:

$$\Pr[\forall x \in \mathcal{V}, \quad \|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2] \geq 1 - \delta$$

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- *"JL for an entire subspace"*

Oblivious Subspace Embedding (OSE)

- **Definition:** a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ is an (ε, δ, d) -oblivious subspace embedding (OSE) if for $\mathbb{R}^n$ if for every given subspace $\mathcal{V} \subset \mathbb{R}^n$ of dimension d , we have:

$$\Pr[\forall x \in \mathcal{V}, \quad \|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2] \geq 1 - \delta$$

- *"JL for an entire subspace"*
- **Equivalent definition:** S is an OSE if for every given matrix $A \in \mathbb{R}^{n \times d}$ we have:

$$\Pr[\forall x \in \mathbb{R}^n, \quad \|SAx\|_2^2 = (1 \pm \varepsilon)\|Ax\|_2^2] \geq 1 - \delta$$

Oblivious Subspace Embedding (OSE)

- **Definition:** a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ is an (ε, δ, d) -oblivious subspace embedding (OSE) if for $\mathbb{R}^n$ if for every given subspace $\mathcal{V} \subset \mathbb{R}^n$ of dimension d , we have:

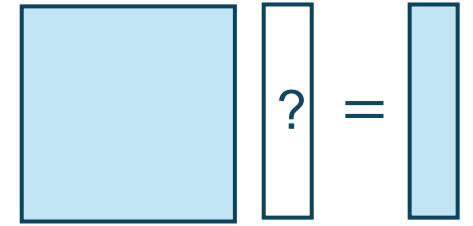
$$\Pr[\forall x \in \mathcal{V}, \quad \|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2] \geq 1 - \delta$$

- *"JL for an entire subspace"*
- **Equivalent definition:** S is an OSE if for every given matrix $A \in \mathbb{R}^{n \times d}$ we have:

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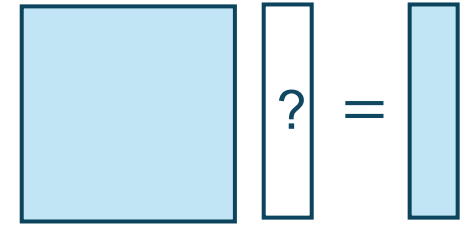
- Why are these the same?

Approximate Regression via OSE



A diagram representing the matrix equation $AX = B$. It consists of a light blue square matrix A , followed by a vertical white rectangle with a black border containing a question mark $?$, an equals sign $=$, and a vertical light blue rectangle representing the vector B .

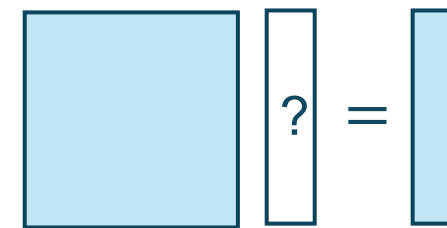
Approximate Regression via OSE



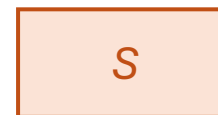
A diagram representing a linear equation. It consists of a light blue square, followed by a vertical rectangle containing a question mark, followed by an equals sign, and finally a light blue vertical rectangle.

- How can we solve ℓ_2 -regression with an OSE?

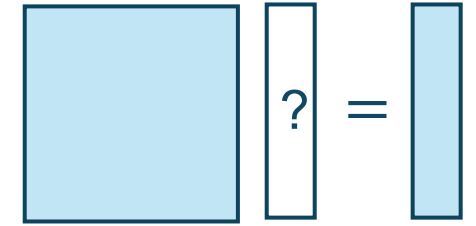
Approximate Regression via OSE


$$\boxed{} \begin{bmatrix} ? \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

- How can we solve ℓ_2 -regression with an OSE?
- Sketch-and-solve:
 - Sample S



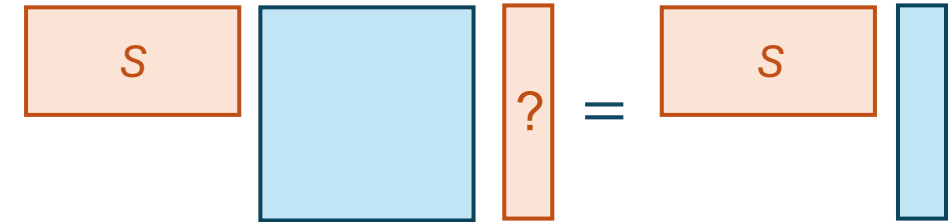
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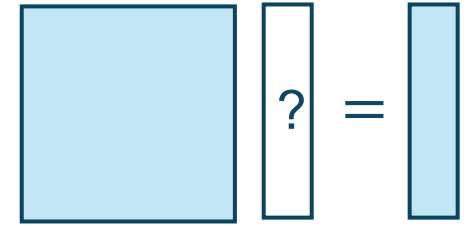
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- Sketch-and-solve:

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$$S \square \begin{bmatrix} ? \end{bmatrix} = S \begin{bmatrix} \end{bmatrix}$$

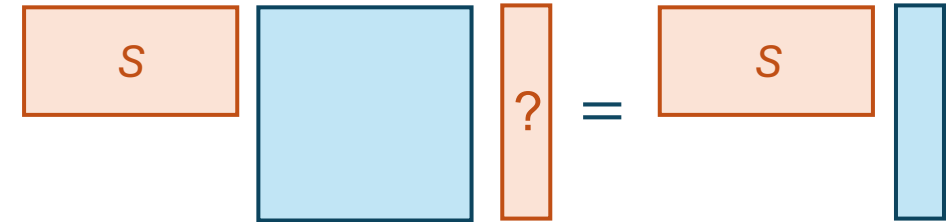
Approximate Regression via OSE


$$A \begin{bmatrix} ? \end{bmatrix} = b$$

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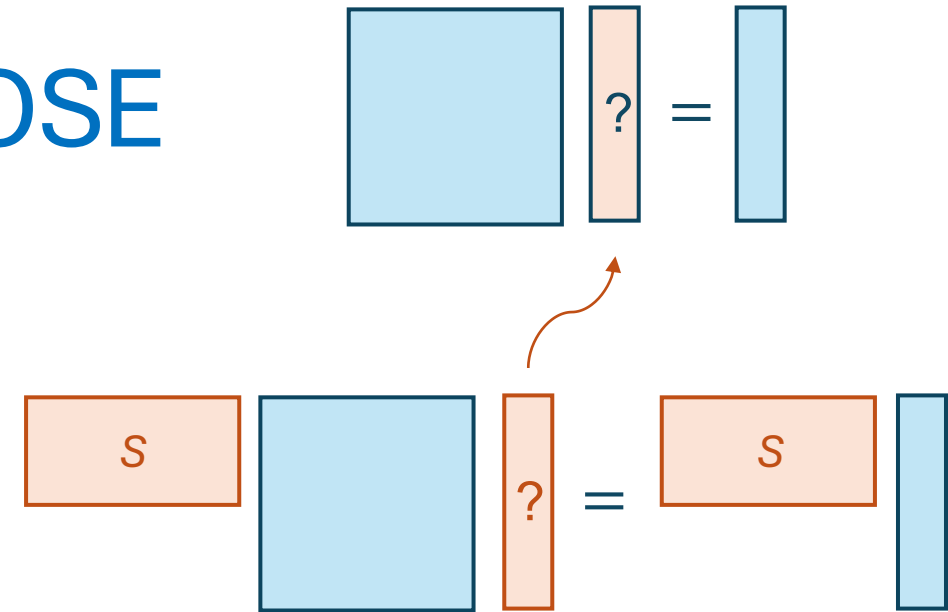
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- Return ???


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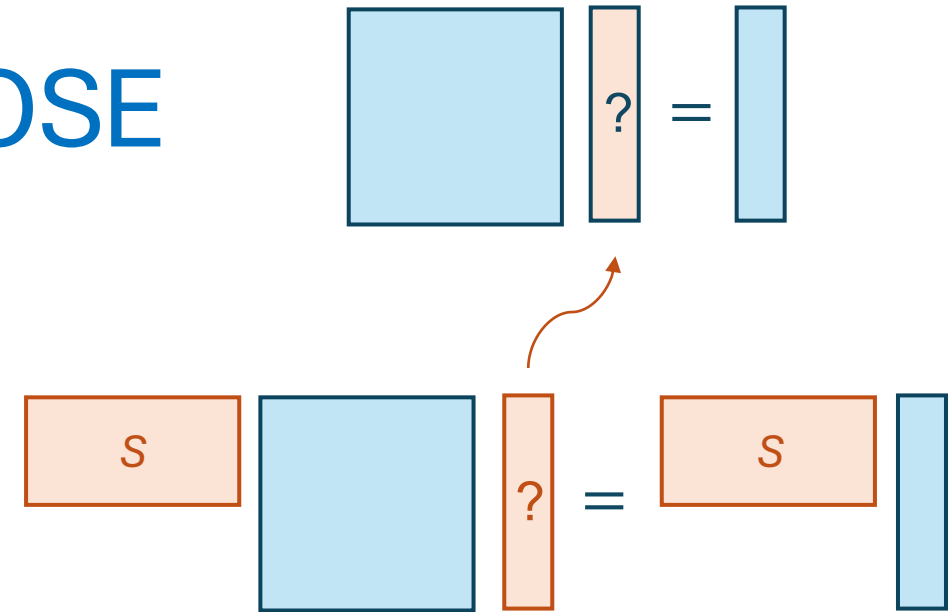
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- **Observe:** if S is an OSE then

$$\|A\tilde{x} - b\|_2 \leq (1 + O(\varepsilon)) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2$$

(why?)



Approximate Regression via OSE

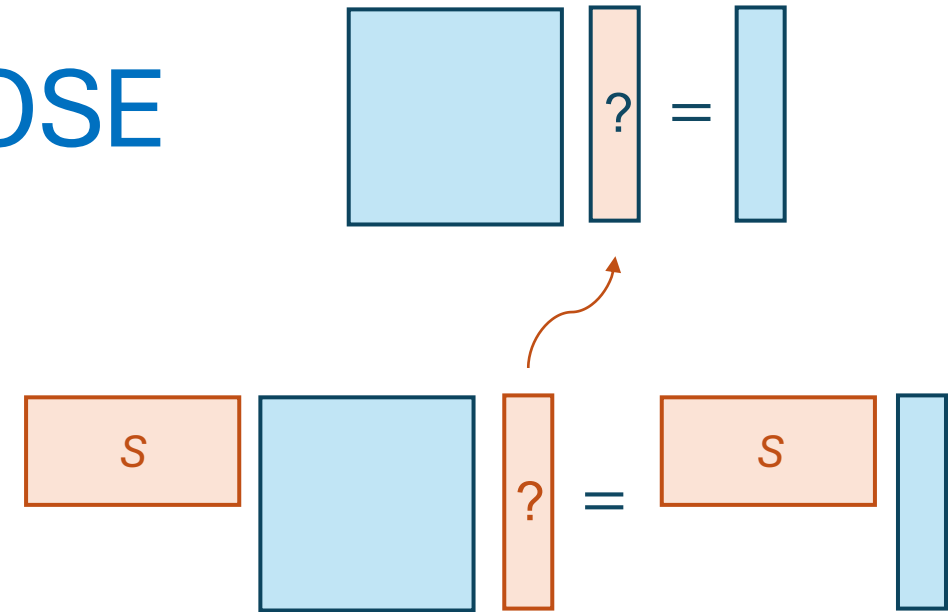
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- **Hence:** Our next goal is to build OSEs (ideas?)



Approximate Regression via JL

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- **Definition (“usual” JL):** a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ satisfies the (ε, δ, N) -JL property if for every given set of N vectors $y_1, \dots, y_N \in \mathbb{R}^n$ we have:

$$\Pr \left[\forall i, j: \|Sy_i - Sy_j\|_2 = (1 \pm \varepsilon) \|y_i - y_j\|_2 \right] \geq 1 - \delta$$

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- **Corollary:** $S \in \mathbb{R}^{m \times n}$ with iid Gaussians and $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$ satisfies (ε, δ, N) -IPJL

OSE via IPJL (1/4)

- Let S be a random matrix satisfying (ε, δ, N) -IPJL
- Let $\mathcal{V} \subset \mathbb{R}^n$ be a subspace of dimension d
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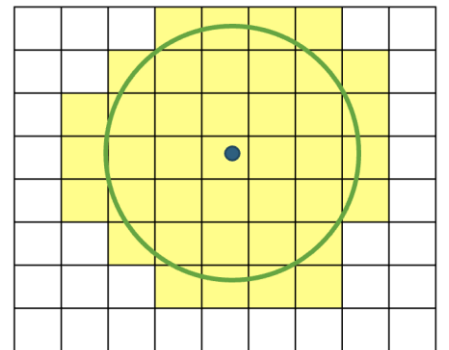
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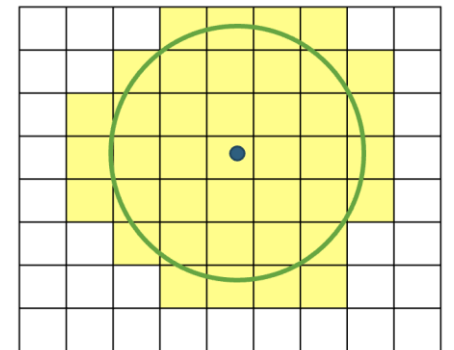
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 - We will take $\eta = 1/2$, so $|Z| = O(1)^d$



OSE via IPJL (2/4)

- Recall: we saw that we have $S \in \mathbb{R}^{m \times n}$ with $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$ satisfies (ε, δ, N) -IPJL
- Recall: Z is a $\frac{1}{2}$ -net of size $O(1)^d$ for the unit sphere in $\mathcal{V}$

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- We apply IPJL to every point in Z

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- Recall: Z is a $\frac{1}{2}$ -net of size $O(1)^d$ for the unit sphere in $\mathcal{V}$
- We apply IPJL to every point in Z
- **Thus:**
 - S has $m = O\left(\frac{d \log(1/\delta)}{\varepsilon^2}\right)$ rows
 - For every $z, z' \in Z$ we have:

$$\Pr[\forall i, j: z^T S^T S z' = z^T z' \pm \varepsilon \|z\|_2 \|z'\|_2] \geq 1 - \delta$$

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 - Take $z_3 \in \frac{1}{4}Z$ such that $\|(x - z_1 - z_2) - z_3\|_2 \leq \frac{1}{8}$

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 - Observe: $\|z_3\|_2 = \frac{1}{4}$
 - We thus have:

$$x = \sum_{i=1}^{\infty} z_i \text{ such that } \forall i, \|z_i\|_2 = \frac{1}{2^i}$$

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- Note: $\sum_{i,j=1}^{\infty} z_i^T z_j = \|x\|_2^2 = 1$, and $\sum_{i,j=1}^{\infty} \|z_i\|_2 \|z_j\|_2 = \sum_{i,j=1}^{\infty} \frac{1}{2^i} \cdot \frac{1}{2^j} = \left(\sum_{i=1}^{\infty} \frac{1}{2^i} \right)^2 = 1$

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- Recall we have IPJL w.p. $1 - \delta$ for all $z, z' \in Z$: $z^T S^T S z' = z^T z' \pm \varepsilon \|z\|_2 \|z'\|_2$
- Therefore, for every unit vectors $x \in \mathcal{V}$:
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- Note: $\sum_{i,j=1}^{\infty} z_i^T z_j = \|x\|_2^2 = 1$, and $\sum_{i,j=1}^{\infty} \|z_i\|_2 \|z_j\|_2 = \sum_{i,j=1}^{\infty} \frac{1}{2^i} \cdot \frac{1}{2^j} = \left(\sum_{i=1}^{\infty} \frac{1}{2^i}\right)^2 = 1$
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OSE via IPJL (4/4)

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- Hence: $\|Sx\|_2^2 = 1 \pm \varepsilon$. 
- **We proved:** JL $\Rightarrow$ IPJL $\Rightarrow$ OSE, thus: every distribution over matrices that satisfies usual JL yields an OSE with $\tilde{O}(d/\varepsilon^2)$ rows.

Approximate Regression via OSE with JL

- **Corollary:**

- Let $A \in \mathbb{R}^{n \times d}$, $b \in \mathbb{R}^d$ be a regression problem.
- Let $m = \tilde{O}(d/\varepsilon^2)$.
- Let $S \in \mathbb{R}^{m \times n}$ be a random matrix with iid Gaussian entries $\frac{1}{\sqrt{m}} N(0,1)$.
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Efficient Linear Algebra: Midway Remarks

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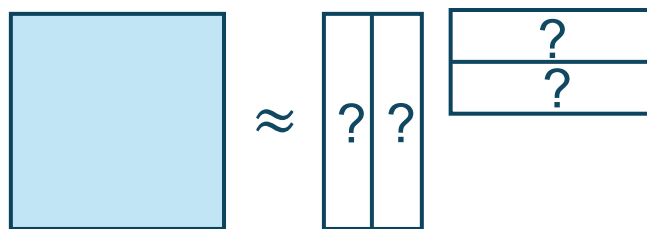
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- We proved them from **generic** JL-like properties (JLMP, IPJL, OSE), thus the proofs work for many JL variants
 - Recall there are many JL variants, including better ones: **Fast JL**, **Sparse JL**, ...
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- Efficient linear algebra goes **beyond JL**:
 - There are non-JL sketching matrices that yield AMM and OSE
 - They lead to even faster solutions for these problems
 - We might see a bit of this in HW3

Approximate Low Rank Approximation



Matrix Low Rank Approximation (LRA) (1/2)

- **Definition** – matrix low-rank approximation:
 - Input: matrix $A \in \mathbb{R}^{n \times d}$ of arbitrary rank, and target rank $k \ll \min\{n, d\}$

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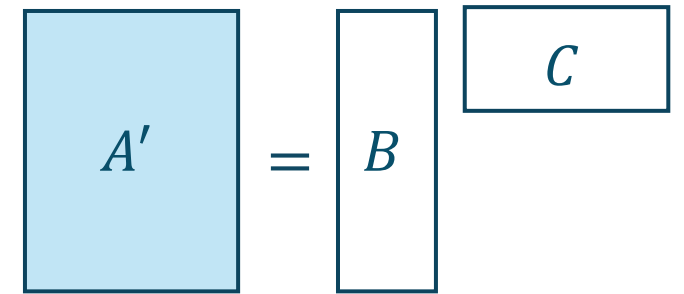
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- **What metric?**

- Common choices: Frobenius norm $\|A - A'\|_F$, spectral norm $\|A - A'\|_2$
- We will focus on Frobenius norm

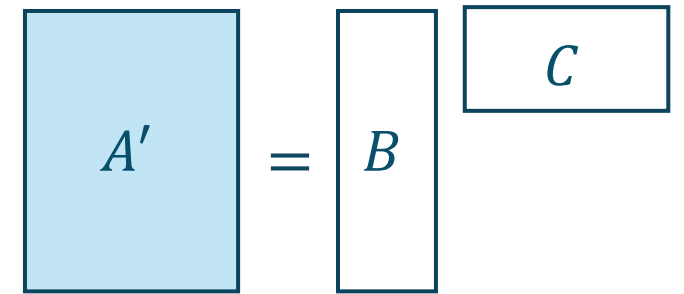
Matrix Low Rank Approximation (LRA) (2/2)

- Every $A' \in \mathbb{R}^{n \times d}$ of rank k can be written as $A' = BC$ where $B \in \mathbb{R}^{n \times k}$ and $C \in \mathbb{R}^{k \times d}$


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A diagram illustrating the matrix equation $A' = BC$. On the left is a light blue square labeled A' . To its right is an equals sign. Further right is a white vertical rectangle labeled B , followed by another equals sign, and finally a white horizontal rectangle labeled C . The rectangles are sized to represent the dimensions of the matrices: B is tall and narrow, C is short and wide, and A' is a square.

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 - How much space it takes to store A vs. B, C ?
 - Given $x \in \mathbb{R}^d$, how much time it takes to compute Ax vs. BCx ?

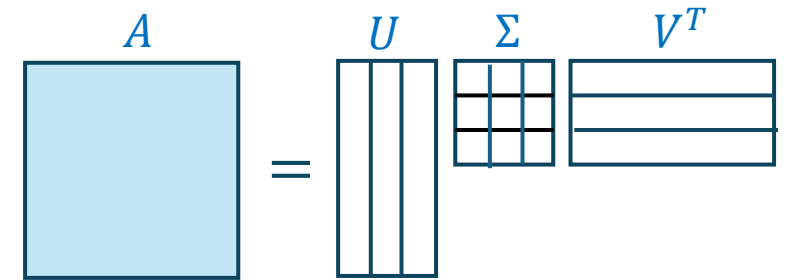
Optimal LRA via SVD (Eckhard-Young-Mirsky Theorem)

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- $U \in \mathbb{R}^{n \times k}$ has orthonormal columns
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- $\Sigma \in \mathbb{R}^{k \times k}$ is diagonal with sorted entries :

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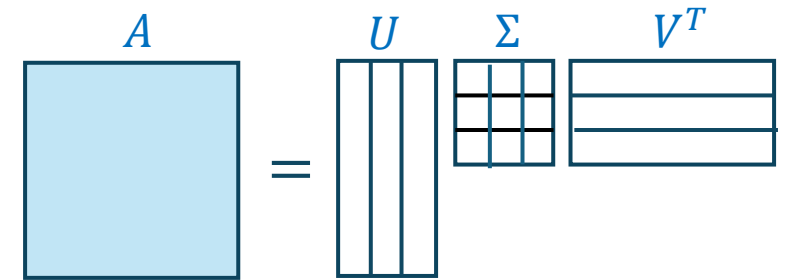
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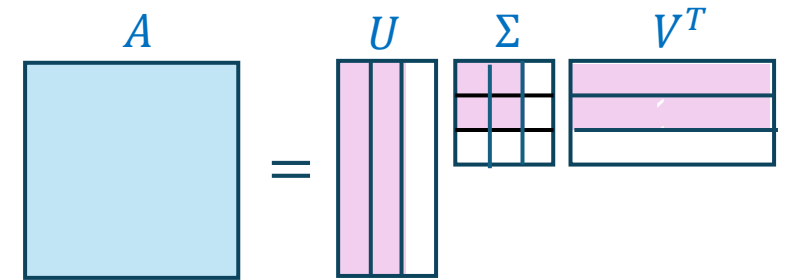
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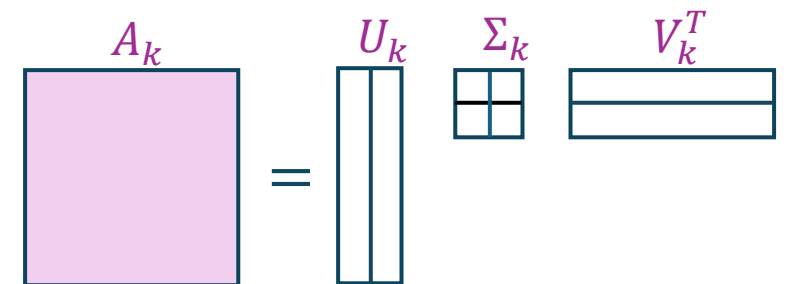
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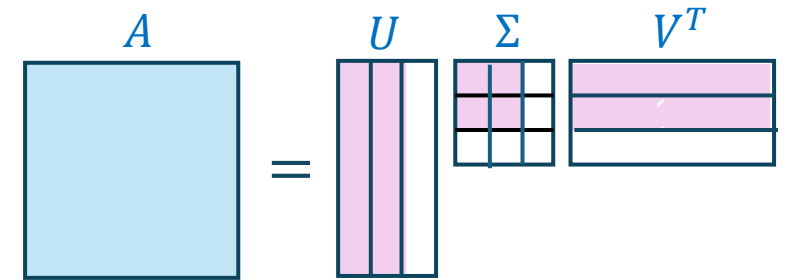


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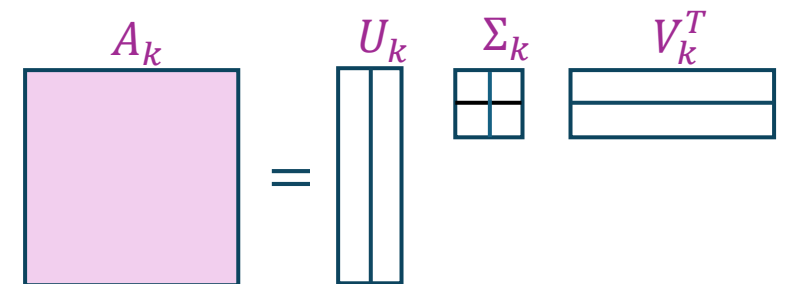
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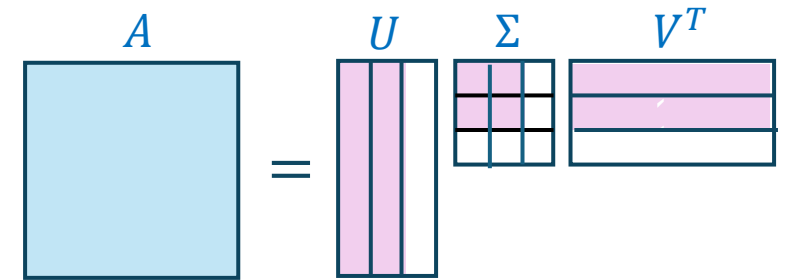
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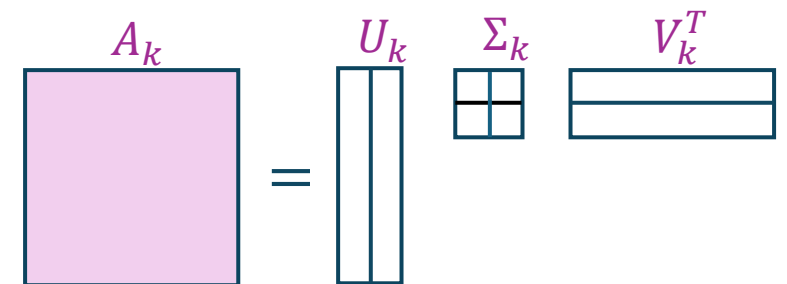
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- **Our goal**: Approx LRA algorithm in time $O(nd \cdot (k/\varepsilon)^{O(1)})$ (linear in the size of A)

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- **Approximate LRA** will choose the best rows from A' from the row-space of SA
 - Intuition: $\dim(SA) = m$
 - Since $m \gg k$, we have enough richness from which to choose rows for A'
 - Since $m \ll n$, choosing the best rows for A' could be efficient

Approximate LRA via Sketching

- **Algorithm “plan”:**

- Sample a sketching matrix $S \in \mathbb{R}^{m \times n}$
- Compute SA
- Compute and return the best rank- k LRA of A within the row-space of SA :

$$A' = \operatorname{argmin}_{A': \operatorname{rows}(A') \subset \operatorname{span}(\operatorname{rows}(SA))} \|A - A'\|_F$$

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- **Prove efficiency:** Given SA , how can we find the best B efficiently?
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Running
time?

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Pythagorean identity



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- Next, we need to show **accuracy**: For a good choice of random sketching matrix S ,
$$\Pr[\|A - [AV]_k V^T\|_F \leq (1 + \varepsilon) \|A - A_k\|_F] \geq 1 - \delta.$$

Questions?

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- Corollary:** (ε, δ) -approximate LRA in time $O(nd \cdot (k/\varepsilon)^{O(1)})$.
 - Linear in the size of A

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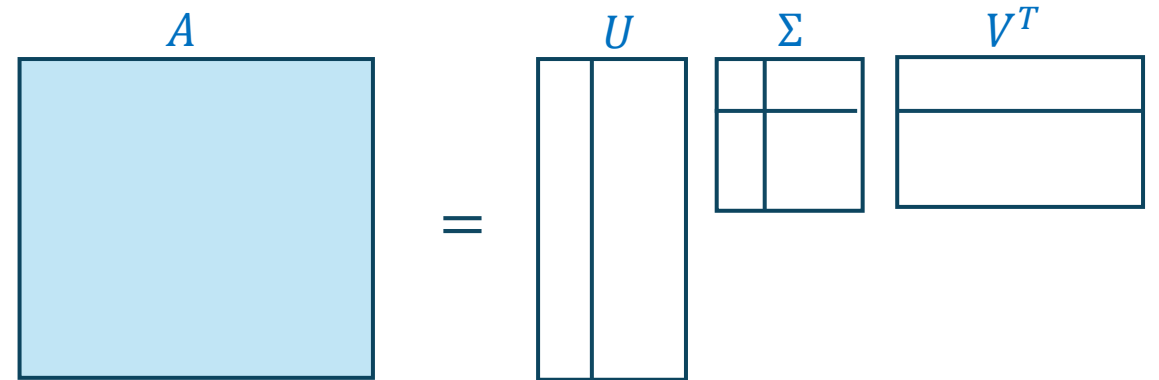
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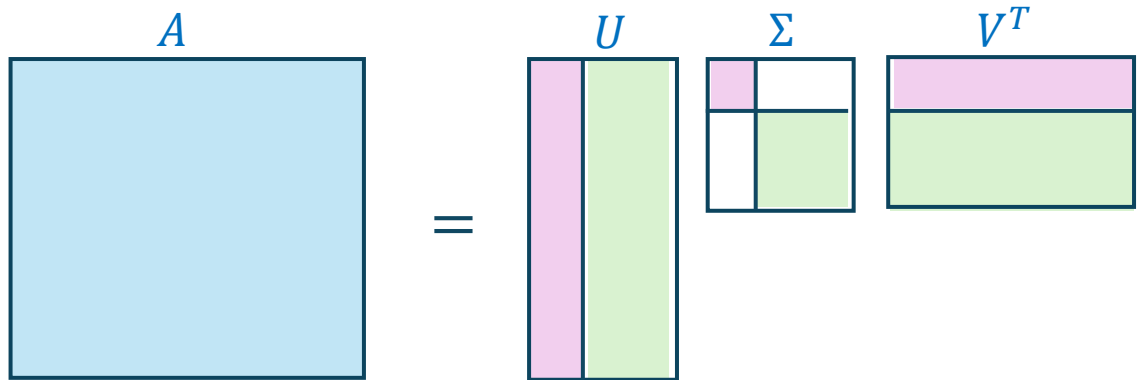
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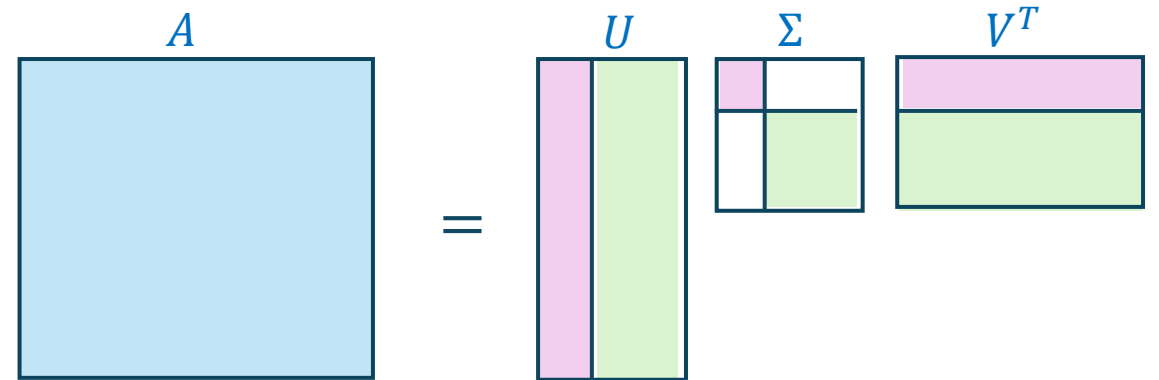
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Observe:

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- We have:
 - $\|BSA - A\|_F^2 = \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2$

LRA Accuracy: Proof

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$$\begin{aligned} \|BSA - A\|_F^2 &= \|U_k Z^{-1} U_k^T S^T S A - A\|_F^2 \\ &= \|\mathbf{U}_k \mathbf{U}_k^T (U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 + \|(I - \mathbf{U}_k \mathbf{U}_k^T)(U_k Z^{-1} U_k^T S^T S A - A)\|_F^2 \end{aligned}$$

???



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Pythagorean
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identity with
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- $= \|U_k Z^{-1} U_k^T S^T SA - A_k\|_F^2 + \|A - A_k\|_F^2 \quad (\text{why?})$

- Therefore, we need to show: $\|U_k Z^{-1} U_k^T S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

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- We have:
 - $\|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 =$

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 - $= \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$

LRA Accuracy: Proof

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 - $= \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$
 - $= \|U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$

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- $= \|U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$
- $= \|U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2$

$$Z^{-1} U_k S^T S U_k = I$$

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- We have:

$$\begin{aligned} \bullet \quad & \|U_k Z^{-1} U_k^T S^T S A - A_k\|_F^2 = \|U_k Z^{-1} U_k S^T S (A_k + A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|U_k Z^{-1} U_k S^T S A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|U_k Z^{-1} U_k S^T S U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|U_k \Sigma_k V_k^T + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \\ \bullet \quad & = \|A_k + U_k Z^{-1} U_k S^T S (A - A_k) - A_k\|_F^2 \end{aligned}$$

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$$Z^{-1} U_k S^T S U_k = I$$

LRA Accuracy: Proof

$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

- We need to show: $\|U_k Z^{-1} S^T A - A_k\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

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- Therefore, we need to show: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2 \varepsilon) \|A - A_k\|_F^2$

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$$\begin{aligned} B &= U_k Z^{-1} U_k^T S^T \\ Z &= U_k^T S^T S U_k \end{aligned}$$

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- We have:
 - $\|U_k Z^{-1} U_k^T S^T S (A - A_k)\|_F^2 \leq \|U_k\|_2^2 \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$

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 - $= k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
- (Why $\|U_k\|_2^2 = k$?) 

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 - $= k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
 - (Why $\|U_k\|_2^2 = k$?) 
- To finish the proof, we will bound both $\|Z^{-1}\|_2^2$ and $\|U_k^T S^T S (A - A_k)\|_F^2$ using AMM

LRA Accuracy: Proof

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- Therefore: $\|U_k^T S^T S(A - A_k) - U_k(A - A_k)\|_F^2 \leq \varepsilon \|U_k\|_F^2 \|A - A_k\|_F^2 = \varepsilon k \|A - A_k\|_F^2$

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- Observe: $U_k(A - A_k) = 0$ (why?)

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LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
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- Recall we assume: S satisfies [AMM](#)
- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4$

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- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$
- Observe: $U_k^T U_k = I$ (why?)

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- Observe: $U_k^T U_k = I$ (why?)
- Therefore: $\|U_k^T S^T S U_k - I\|_F^2 \leq \varepsilon k^2$

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- Therefore: $\|U_k^T S^T S U_k - I\|_F^2 \leq \varepsilon k^2$
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- Therefore: $\|U_k^T S^T S U_k - U_k^T U_k\|_F^2 \leq \varepsilon \|U_k\|_F^4 = \varepsilon k^2$
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- Therefore: $\|U_k^T S^T S U_k - I\|_F^2 \leq \varepsilon k^2$
- Therefore: $\|U_k^T S^T S U_k - I\|_2^2 \leq \varepsilon k^2$ (why?) (note we moved from Frobenius to spectral norm)
- We choose ε small enough so that $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$

LRA Accuracy: Proof

- Recall: $Z = U_k^T S^T S U_k$
- We need to bound: $\|Z^{-1}\|_2^2$ (and, en route, finally prove that Z is invertible)
- We showed: $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$

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- **Lemma:** If a square matrix M satisfies $\|M - I\|_2 \leq \alpha$ for $\alpha < 1$, then all of the eigenvalues of M are in $[1 - \alpha, 1 + \alpha]$. (Proof?)

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- **Lemma:** If a square matrix M satisfies $\|M - I\|_2 \leq \alpha$ for $\alpha < 1$, then all of the eigenvalues of M are in $[1 - \alpha, 1 + \alpha]$. (Proof?)
- **Corollary:** All eigenvalues of Z are between $2/3$ and $4/3$.

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- Recall: $Z = U_k^T S^T S U_k$
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- **Lemma:** If a square matrix M satisfies $\|M - I\|_2 \leq \alpha$ for $\alpha < 1$, then all of the eigenvalues of M are in $[1 - \alpha, 1 + \alpha]$. (Proof?)
- **Corollary:** All eigenvalues of Z are between $2/3$ and $4/3$.
- In particular:
 - Z is invertible.
 - $\|Z^{-1}\|_2^2 \leq \frac{16}{9} = O(1)$.

LRA Accuracy: End of Proof (Finally!)

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- Our goal was to prove: $\|BSA - A\|_F^2 \leq (1 + O(k^2\varepsilon))\|A - A_k\|_F^2$

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- We showed it's enough to prove: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$

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- We proved:
 - $\|U_k Z^{-1} U_k S^T S(A - A_k)\|_F^2 \leq k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S(A - A_k)\|_F^2$
 - $\|U_k^T S^T S(A - A_k)\|_F^2 \leq \varepsilon k \|A - A_k\|_F^2$
 - $\|Z^{-1}\|_2^2 = O(1)$

LRA Accuracy: End of Proof (Finally!)

- Our goal was to prove: $\|BSA - A\|_F^2 \leq (1 + O(k^2\varepsilon))\|A - A_k\|_F^2$
- We showed it's enough to prove: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$
- We proved:
 - $\|U_k Z^{-1} U_k S^T S(A - A_k)\|_F^2 \leq k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S(A - A_k)\|_F^2$
 - $\|U_k^T S^T S(A - A_k)\|_F^2 \leq \varepsilon k \|A - A_k\|_F^2$
 - $\|Z^{-1}\|_2^2 = O(1)$
- Which together yields: $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$ as needed.

